

Midterm Exam

CS 101 Algorithmic Problem Solving

Fall 2023

Name(s): _____

HU ID (*e.g., xy01042*): _____

Exam Instructions

- **Exam Duration:** 2 hours and 30 minutes
- **Number of Questions:** This exam contains 4 coding questions and 1 debugging question.
- **Technical Requirements:** You are not allowed to use code editors, compilers, or online resources during this exam.
- **Communication:** During the exam, do not communicate with anyone except the instructor or exam proctors if you have questions about the exam.
- **Honesty:** Academic honesty is expected. Do not engage in any form of cheating, plagiarism, or unauthorized collaboration.
- **Answer Format:** Write your answers clearly in the provided spaces.
- **Time Management:** Monitor the time carefully and pace yourself. Do not spend too much time on any one question or section.
- **Exam Completion:** Before submitting, take a moment to review your answers to ensure you have completed all questions and that your responses are clear and accurate.

Good luck!

1. Nim Game

In the tranquil town of Blacksburg, Izzah and Haisum, renowned for their keen intellect, found themselves engrossed in the Nim Game. Given a heap of stones represented by an integer n , both players take turns removing 1 to 3 stones from the heap. The player who takes the last stone wins the game. Izzah plays first.

Write a Python function called `can_win(n)` that takes an integer n as input and returns True if Izzah can win the Nim Game assuming both players play optimally, and False otherwise.

Constraints

- $1 \leq n \leq 2^{31} - 1$

Interaction

The input comprises an integer n denoting the number of stones.

The output must be True if Izzah can win the Nim Game assuming both players play optimally, and False otherwise.

Sample

Input	Output
5	True
6	False

In the first case, Izzah can remove 1 stone. No matter how many stones Haisum removes (1, 2, or 3), Izzah can always remove the remaining stones and win the game.

In the second case, no matter how many stones Izzah removes (1, 2, or 3), Haisum can always mirror the move and take the last stone, winning the game.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
10	
25	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

2. Character Identification

Computers save all characters as numbers. There is a standard code called ASCII that they use to correspond a character with a number. In Python, the ASCII code of a character can be found using the built-in function, `ord`. For example,

```
>>> ord('4')
52
>>> ord('X')
88
>>> ord('h')
104
```

The ASCII codes of numbers and letters are as follows.

Characters	Codes
Upper case letters, A-Z	65-90
Lower case letters, a-z	97-122
Digits, 0-9	48-57
Special characters	all other codes

Your task is to write a function which takes character C as an argument, and prints whether the character is a lower case letter, upper case letter, digit, or special character.

Constraints:

- $C \in \text{Uppercase Letters } (A-Z) = \{A, B, C, \dots, Z\}$
- $C \in \text{Lowercase Letters } (a-z) = \{a, b, c, \dots, z\}$
- $C \in \text{Digits } (0-9) = \{0, 1, 2, \dots, 9\}$
- $C \in \text{Special Characters} = \{\text{characters other than } A-Z, a-z, \text{ and } 0-9\}$

Interaction

The input consists of a single line which contains a string C of length 1.

The output must print whether C is a lower case letter, upper case letter, digit, or special character.

Sample

Input	Output
B	B is an uppercase letter
i	i is a lower case letter
*	* is a special character

In the first case, The ASCII code for B is 66, which is between 65 and 90.

In the second case, The ASCII code for i is 105, which is between 97 and 122.

In the third case, The ASCII code for * is 42, which falls in none of the given ranges.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
5	
&	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

3. Simple symbols

You are working as a software developer at Habib University. The university's computer science department is organizing a programming competition for its students. As a part of the competition, you are tasked to create a function called **SimpleSymbols(s)**.

Habib University takes pride in fostering creativity and innovation among its students. In this context, the function **SimpleSymbols(s)** is crucial to ensure the integrity of the competition. The function needs to determine whether the input sequence provided by the participants is acceptable.

The input sequence, denoted as s , is a combination of $+$ and $=$ symbols, with one or more numbers sandwiched between them. An acceptable sequence is defined as one where:

- Each number in s is surrounded by exactly one $+$ symbol on each side. For example, $+2+$, $+5+$, and $+1+$ are valid patterns, while $++2++$, $=5=$, and $==1+$ are invalid patterns.
- There should not be more than one consecutive $+$ or $=$ symbol in the input sequence. For example, $++$ and $==$ are invalid patterns.

Your task is to create the `SimpleSymbols(s)` function, adhering to Habib University's standards of precision and clarity. If the input sequence meets the criteria, the function should return `True`. Otherwise, it should return `False`.

Constraints

- $\text{len}(s) \geq 1$ is *True*
- $S \in \{N, +, =\}$

Interaction

The input comprises of a string s .

The output must contain 'True' or 'False' based on the specified condition.

Sample

Input	Output
$++5+==+2++==2$	False
$+1+=+=+6+=+9+=+1+=+=+5+$	True

In the first case, the number 2 is not surrounded by plus symbols, and the plus sign appears in a contiguous location.

In the second case, all numbers are enclosed by $+$ symbols, making the sequence acceptable for the competition.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
$=+2+=+2+=+=+2+=+2+=+=+2+=$	
$3++7+$	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

4. Word length puzzle

Sara has a strange habit. Whenever someone says a sentence, she is stuck on the last word, trying to count the length of the last word they said. This habit is making her lose her mind.

Write a function *last_word(sentence)*, that computes the length of the last word in a sentence uttered by someone during a conversation with Sara, and returns it to her ear (in this case, the screen) when she calls the function. A word, in this context, is a sequence of characters without spaces.

Make sure to call the main function to print the output.

Constraints:

- $1 \leq \text{sentence.length} \leq 1000$

- *sentence* consists of only English letters and spaces ' '.
- There will be at least one word in *sentence*.

Interaction

Input comprises of a string *sentence*.

Output must be an integer representing the length of the last word in that sentence.

Sample

Input	Output
"Hello World"	5
"this midterm is a piece of cake"	4

In the first case, the length of the last word "World" is returned, which is 5.

In the second case, the length of the last word "cake" is returned, which is 4.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
"luffy is still joyboy"	
"fly me to the moon"	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

5. Floyd's Triangle

Given a positive integer n , your task is to write a function $Floyd(n)$, which prints the first n rows of Floyd's triangle.

Constraints

- $1 \leq n \leq 100$

Interaction

The input comprises a single line containing an integer n .

The output must contain the first n rows of Floyd's triangle indented to the right.

Sample

Input	Expected Output
3	1 2 3 4 5 6
5	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

In the first case, a right aligned floyd's triangle is created with 3 rows.

In the first case, a right aligned floyd's triangle is created with 5 rows.

Proposed Solution

```
def Floyd(n):  
    num = 0  
    for i in range(n):
```

```
    for j in range (i):  
        print (i, end = ' ' )  
        num +=1  
    print()
```

Floyd(n)

Dry Run

Using the inputs provided in the Sample section above, dry run the proposed pseudocode in the space below to find out errors.

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

Rough Work